Drug Interaction Severity Prediction

Generative AI provided limited assistance in line with the course policy on usage of AI.

Introduction

Modern medication management can involve patients taking five or more medications simultaneously, a practice referred to as polypharmacy [1]. Physicians account for drug-drug interactions (DDIs) when prescribing and often use software to flag potential interactions [1]. Supplementary screening approaches could potentially improve patient safety and help providers quickly identify DDIs [1].

DDIs represent a significant clinical risk of polypharmacy. Data from 2015 to 2018 showed roughly one in four American adults filled three or more prescriptions monthly, climbing above two-thirds in patients aged 65 and older [1]. Federal regulators have revised DDI safety guidance since the late 1990s in response to documented efficacy and safety failures [1]. Clinical trial testing cannot cover every possible drug combination, leaving pharmacovigilance systems to detect harmful interactions reactively, often after patient harm has occurred.

This project uses the Deep-Pharma Multimodal Intelligence DataHack 3.0 Kaggle competition [2], selected for its direct relevance to biomedical and health informatics, its use of real FDA FAERS data, and the challenge of a cold-start test set of unseen molecular scaffolds across three simultaneous prediction tasks. The model predicts interaction severity (Minor, Moderate, or Major), which of 50 adverse events will occur, and per-adverse-event risk using Proportional Reporting Ratio (PRR). Success is evaluated using a composite Hardcore Clinical Score from 0.0 to 1.0, weighted across severity classification (40%, Macro F1), side effect prediction (30%, Micro F1), and PRR regression (30%, inverse RMSE). A reliable model has direct implications for patient safety, offering a scalable screening layer that could strengthen clinical decision support at the point of prescribing.

This project addresses two research questions. RQ1: Can a multi-task machine learning model accurately predict DDI severity, adverse event occurrence, and PRR risk from pharmacological and molecular drug pair features? RQ2: What natural groupings exist within drug pair interaction profiles, and do these clusters align with meaningful differences in severity and side effect patterns? RQ2 extends beyond the competition scope as an exploratory contribution applying unsupervised learning to examine whether adverse event profiles segment in clinically meaningful ways.

Methods

Data

The training dataset contained 15,284 drug pair observations across 126 columns (Table A1). The severity target was multiclass: Moderate (69.1%), Major (26.7%), and Minor (4.1%), yielding a 16.8:1 imbalance ratio between the majority and minority classes (Table B1, Figure C1). This imbalance directly motivated Macro F1 as the primary severity evaluation metric, which weights all classes equally. Adverse event burden was highly skewed: 68.5% of pairs reported no adverse events and 14.7% reported 40 or more, producing a bimodal distribution (Table B2, Figure C2).

Among non-zero PRR entries, values ranged up to 240.0 with a mean of 3.89 and median of 2.50 (Table B3, Figure C4). Only 21.5% of PRR entries were non-zero. CYP450 enzyme profiles were missing for 59.5% of Drug A and 47.1% of Drug B observations; FDA warning fields were missing in 31.0% and 11.3% of cases respectively (Table B4, Figure C5). Missing values were replaced with empty strings prior to vectorization so absent fields contributed no signal.

Preprocessing

All non-target text columns including mechanism of action, pharmacodynamics, ADME fields, CYP450 profiles, indications, warnings, and toxicity data were concatenated into a single string per drug pair. An 80/20 stratified train/test split was applied before feature extraction. TF-IDF vectorization with 2,000 tokens and English stop word removal was fit on the training split only to prevent leakage. SMILES strings were included as raw text; this captures token-level molecular notation rather than true structural properties and likely limits PRR performance. All three tasks used the same feature matrix.

Severity Classification

Three classifiers were trained: Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression served as the interpretable baseline. Random Forest reduces variance through bootstrap sampling and handles text-derived features without scaling. Gradient Boosting fits each new tree to residual errors of the current ensemble, learning feature interactions the other models cannot capture. All three were evaluated using 5-fold cross-validation on a 3,000-observation training subset for runtime efficiency, then scored on the held-out test set using multiclass AUC and Macro F1 (Table B6, Figure C8). Final hyperparameters: Logistic Regression `max_iter=1000`; Random Forest `n_estimators=100`; Gradient Boosting `n_estimators=100`, `max_depth=3`, `learning_rate=0.1`. Gradient Boosting was selected based on the highest AUC (0.9252) and Macro F1 (0.5666) on the test set (Table B5, Figure C6). Feature importance was extracted via MDI. A shallow decision tree (max depth=3) was fit separately as an interpretability tool only (Figures C9, C10).

Binary Side Effect Prediction

A MultiOutputClassifier wrapping Logistic Regression was fit independently per label on the same TF-IDF matrix. Performance was evaluated using global Micro F1, which penalizes null predictions and requires active detection of true positives, and per-label F1 and AUC across all 50 adverse events on the held-out test set (Table B11, Table B12, Figure C14).

PRR Regression

A multi-output Ridge regressor was trained to predict PRR for each of the 50 adverse events. Predictions were clipped at zero. RMSE was computed only on pairs where ground truth PRR was greater than zero, penalizing errors on confirmed signal pairs. Inverse RMSE (1 / 1 + RMSE) was used as the score component, where higher values indicate better performance (Table B13, Table B14, Figure C15). Ridge was selected for stability on high-dimensional sparse input.

Unsupervised Clustering

To address RQ2, K-means clustering was applied to the 50 binary adverse event columns. A silhouette score sweep across k=2 through k=6 selected k=2 with a score of 0.776 (Figure C11). Severity label composition and adverse event prevalence were examined within each cluster to assess clinical alignment (Figures C12, C13). Full reproducibility details and source code are available in the project GitHub repository.

Results and Evaluation

Micro-average AUC was the primary severity metric. Micro F1 was used for side effect prediction. Inverse RMSE on masked data was used for PRR. These three metrics compose the Hardcore Clinical Score weighted as severity (40%), side effects (30%), and PRR (30%).

Severity Classification

All three models achieved AUC above 0.91, indicating strong class-level discrimination (Table B7, Figure C6). Gradient Boosting was highest across all metrics and selected as the final submission model. AUC differences are narrow (delta less than 0.008), suggesting further gains require feature engineering rather than model switching.

Model	Accuracy	Macro F1	AUC
Logistic Regression	0.7694	0.5395	0.9241
Random Forest	0.7661	0.5639	0.9146
Gradient Boosting	0.7746	0.5666	0.9252

Cross-validation accuracy was stable across all models ranging from 0.741 to 0.757 (Table B6, Figure C8). Per-class recall showed a consistent failure pattern across all models (Figure C7, Tables B8-B10). Moderate recall exceeded 91% in Gradient Boosting. Major recall ranged from 42% to 50%. Minor peaked at 19.8%. The 16.8:1 imbalance drives this failure and confirms Macro F1 as the appropriate metric. Feature importance identified lymphocytes as the dominant token, followed by smooth, exercise, diet, and glucose — consistent with known high-risk DDI drug classes affecting metabolic and immunological pathways (Figures C9, C10).

Binary Side Effect Prediction

The multi-label classifier achieved Micro F1 of 0.5772, Macro F1 of 0.5698, and mean per-label AUC of 0.8938 across all 50 adverse events (Table B11). Per-label F1 ranged from 0.499 for Condition Aggravated to 0.666 for Dyspnoea with a mean of 0.570 (Table B12, Figure C14). High-prevalence systemic events including Dyspnoea, Nausea, Fatigue, and Diarrhoea were predicted most reliably. Rare events including Condition Aggravated, Loss of Consciousness, and Myocardial Infarction showed the weakest F1. Mean AUC of 0.8938 indicates strong ranking ability despite modest F1.

Metric	Value
Micro F1	0.5772
Macro F1	0.5698
Mean per-label AUC	0.8938

PRR Regression

PRR regression was the weakest task. Mean masked RMSE was 4.576 with mean inverse RMSE of 0.2023 (Table B13). Per-label inverse RMSE ranged from 0.088 for Atrial Fibrillation to 0.440 for Drug Ineffective (Table B14, Figure C15). Events with low PRR variance including Drug Ineffective, Death, and Nausea were predicted most accurately. Events with high variance including Atrial Fibrillation, Renal Failure Acute, and Drug Interaction showed the largest errors.

Metric	Value
Mean Masked RMSE	4.576
Mean Inverse RMSE	0.2023

Composite Score

The estimated composite Hardcore Clinical Score was 0.4605. The Kaggle submitted score was 0.5134, generated using Gradient Boosting for severity combined with the multi-label and Ridge regression outputs (Table B15). PRR regression contributed the lowest component score.

Component	Value
Severity Macro F1 (x0.40)	0.5666
Side Effect Micro F1 (x0.30)	0.5772
PRR Inverse RMSE (x0.30)	0.2023
Estimated Composite	0.4605
Kaggle Submitted Score	0.5134

Unsupervised Clustering

K-means identified k=2 as optimal with silhouette score 0.776, declining monotonically through k=6 (Figure C11). Cluster 1 had uniformly high adverse event prevalence (0.84 to 0.97) and Cluster 2 near-zero prevalence (0.02 to 0.05) across all 20 differentiating events (Figure C12). This separation is consistent with the bimodal adverse event distribution. Severity composition was near-identical: Moderate comprised 75.1% of Cluster 1 and 67.2% of Cluster 2 (Figure C13). This directly answers RQ2. Adverse event burden and interaction severity are independent dimensions. The clusters reflect high-symptom versus low-symptom drug pair profiles rather than a clinically meaningful severity stratification.

Discussion

Pharmacological text features from FDA FAERS data carry meaningful predictive signal for DDI modeling. TF-IDF representations produced strong discriminative performance for severity classification and binary side effect prediction but were insufficient for precise PRR quantification. The test set consisted of 3,974 held-out pairs with new molecular scaffolds, making generalization harder than a standard random split.

Severity classification and binary side effect prediction achieved AUC above 0.91 and 0.89 respectively, confirming text-derived features capture clinically relevant interaction patterns across unseen scaffolds. Gradient Boosting was the strongest classifier. Dominant tokens including lymphocytes, glucose, and prolongation are

pharmacologically interpretable and consistent with known high-risk DDI drug classes. Per-class recall for Major and Minor remained poor, driven by the 16.8:1 imbalance. PRR regression produced a mean inverse RMSE of 0.2023, indicating TF-IDF lacks the structural resolution needed for continuous pharmacovigilance signal estimation. The composite Hardcore Clinical Score of 0.4605 reflects this unevenness across the three tasks.

K-means clustering identified two drug pair profiles defined by adverse event burden rather than severity. High-burden and low-burden pairs separated cleanly with a silhouette score of 0.776, but severity composition was near-identical within each cluster. Adverse event burden and interaction severity are independent dimensions in this dataset. This directly answers RQ2 with a meaningful negative result. A follow-up study incorporating CYP450 overlap features and molecular fingerprints could test whether structural pharmacological features recover clusters that align with severity.

Limitations

CYP450 profiles were missing in 47 to 60% of observations and contributed no structured signal, representing the most significant constraint on model performance. SMILES-derived molecular fingerprints were not utilized, limiting PRR performance and generalization to new scaffolds. The cross-validation subset of 3,000 observations may not fully represent the full training distribution. The binary classifier treats all 50 labels independently, ignoring co-occurrence structure. The cold-start test set structurally disadvantages text-only models and would require molecular fingerprinting or graph-based drug representations to fully address.

Appendix

A. Feature Dictionary

Category	Column(s)	Description
Identifiers	`Pair_ID`	Unique row identifier for each drug interaction pair
	`Drug_A_Name`, `Drug_B_Name`	Generic INN names for each drug
	`SMILES_A`, `SMILES_B`	Text-encoded 2D molecular structure and topology
Mechanism	`Mechanism_A/B`	Molecular targets (receptors, enzymes) activated or inhibited
	`Pharmacodynamics_A/B`	Biochemical and physiological effects on the body
Pharmacokinetics	`Absorption_A/B`	Bioavailability and absorption rate details

Category	Column(s)	Description
	`Metabolism_A/B`	Site and pathway of metabolic breakdown
	`Elimination_Route_A/B`	Exit pathway from the body (renal, biliary, etc.)
	`Half_Life_A/B`	Time for plasma concentration to decrease by 50%
	`Protein_Binding_A/B`	Fraction of drug bound to plasma proteins (%)
Metabolic Enzymes	`CYP450_Enzymes_A/B`	CYP450 isozymes involved in metabolism; overlap is a key DDI predictor
Clinical Profile	`Indication_A/B`	Approved therapeutic indications
	`Warning_A/B`	FDA Black Box warnings and safety precautions
	`Toxicity_A/B`	Toxic effects, LD50 values, and overdose profiles
Target: Severity	`Severity`	Interaction severity class — Major, Moderate, or Minor
Target: Side Effects	`Target_Binary_[SideEffect]` x 50	Multi-label binary flags; 1 = side effect present, 0 = absent
Target: PRR Risk	`Target_PRR_[SideEffect]` x 50	Proportional Reporting Ratio (float); >0 = statistical signal detected

B. Data Tables

Table B1. Severity Class Distribution

Severity	Count	Percent (%)	Imbalance Ratio
Moderate	10,568	69.1	16.8
Major	4,086	26.7	6.5
Minor	630	4.1	1.0

Table B2. Adverse Event Burden per Drug Pair

Statistic	Value
Mean AEs per pair	10.77
Median AEs per pair	0.00
Min AEs	0
Max AEs	50
Pairs with 0 AEs (%)	68.5
Pairs with 40+ AEs (%)	14.7

Table B3. PRR Distribution Summary

Statistic	Value
Total PRR entries	764,200
Non-zero PRR entries (%)	21.5
Mean PRR (non-zero)	3.892
Median PRR (non-zero)	2.500
Max PRR	240.000
PRR columns	50

Table B4. Feature Missingness Rate

Feature	Missing (%)
CYP450_Enzymes_A	59.5
CYP450_Enzymes_B	47.1
Warning_A	31.0
Indication_A	19.1
Indication_B	11.3
Warning_B	11.3

Table B5. Model Performance Summary — Severity Classification (Test Set)

Model	Accuracy	Macro F1	AUC
Logistic Regression	0.7694	0.5395	0.9241
Random Forest	0.7661	0.5639	0.9146
Gradient Boosting	0.7746	0.5666	0.9252

Table B6. 5-Fold Cross-Validation Accuracy (Training Subset)

Model	CV Mean	CV Std
Logistic Regression	0.7537	0.0083
Random Forest	0.7410	0.0152

Model	CV Mean	CV Std
Gradient Boosting	0.7570	0.0153

Table B7. ROC AUC by Model (Test Set)

Model	AUC
Logistic Regression	0.9241
Random Forest	0.9146
Gradient Boosting	0.9252

Table B8. Confusion Matrix — Logistic Regression (Test Set)

Actual / Predicted	Major	Minor	Moderate
Major	402	0	415
Minor	4	14	108
Moderate	176	2	1,936

Table B9. Confusion Matrix — Random Forest (Test Set)

Actual / Predicted	Major	Minor	Moderate
Major	408	2	407
Minor	9	23	94
Moderate	185	18	1,911

Table B10. Confusion Matrix — Gradient Boosting (Test Set)

Actual / Predicted	Major	Minor	Moderate
Major	342	1	474
Minor	5	25	96
Moderate	101	12	2,001

Table B11. Binary Side Effect Prediction Summary

Metric	Value
Micro F1	0.5772
Macro F1	0.5698
Mean per-label AUC	0.8938

Table B12. Binary Side Effect Prediction — Per-Label F1 and AUC

Adverse Event	F1	AUC
Dyspnoea	0.666	0.897
Nausea	0.662	0.887

Adverse Event	F1	AUC
Fatigue	0.651	0.894
Diarrhoea	0.646	0.890
Asthenia	0.635	0.899
Vomiting	0.633	0.890
Pyrexia	0.631	0.890
Pneumonia	0.628	0.893
Pain	0.626	0.891
Anaemia	0.616	0.897
Headache	0.609	0.891
Fall	0.609	0.902
Dizziness	0.608	0.898
Anxiety	0.601	0.901
Hypertension	0.590	0.897
Depression	0.588	0.902
Weight decreased	0.583	0.892
Arthralgia	0.582	0.891
Oedema peripheral	0.578	0.889
Pain in extremity	0.574	0.888
Abdominal pain	0.571	0.894
Back pain	0.566	0.892
Drug ineffective	0.564	0.885
Malaise	0.564	0.883
Chest pain	0.564	0.890
Insomnia	0.563	0.899
Death	0.563	0.894
Hypotension	0.562	0.896
Rash	0.554	0.891
Decreased appetite	0.553	0.894
Renal failure	0.551	0.902
Renal failure acute	0.550	0.886
Confusional state	0.546	0.899
Cough	0.547	0.888
Somnolence	0.545	0.901
Dehydration	0.543	0.893
Constipation	0.538	0.899
Urinary tract infection	0.533	0.897
Gait disturbance	0.533	0.896
Atrial fibrillation	0.527	0.896
General physical health deterioration	0.526	0.902
Haemoglobin decreased	0.522	0.890
Drug interaction	0.521	0.890

Adverse Event	F1	AUC
Hypoaesthesia	0.521	0.901
Weight increased	0.518	0.896
Pruritus	0.515	0.889
Tremor	0.508	0.897
Myocardial infarction	0.504	0.893
Loss of consciousness	0.502	0.885
Condition aggravated	0.499	0.892

Table B13. PRR Regression Summary

Metric	Value
Mean Masked RMSE	4.576
Mean Inverse RMSE (score component)	0.2023

Table B14. PRR Regression — Per-Label RMSE and Inverse RMSE

Adverse Event	RMSE (masked)	Inverse RMSE
Drug ineffective	1.271	0.4404
Death	1.397	0.4172
Nausea	1.800	0.3571
Fatigue	1.925	0.3419
Rash	1.984	0.3351
Dyspnoea	2.190	0.3134
Malaise	2.729	0.2682
Pain in extremity	2.745	0.2670
Pain	2.838	0.2606
Insomnia	2.963	0.2523
Diarrhoea	3.033	0.2480
Headache	3.065	0.2460
Arthralgia	3.140	0.2416
Dizziness	3.358	0.2295
Gait disturbance	3.683	0.2135
Loss of consciousness	3.714	0.2121
Pruritus	3.773	0.2095
Vomiting	3.789	0.2088
Weight decreased	3.871	0.2053
Confusional state	4.058	0.1977
Myocardial infarction	4.063	0.1975
Chest pain	4.069	0.1973
Condition aggravated	4.206	0.1921
Depression	4.334	0.1875

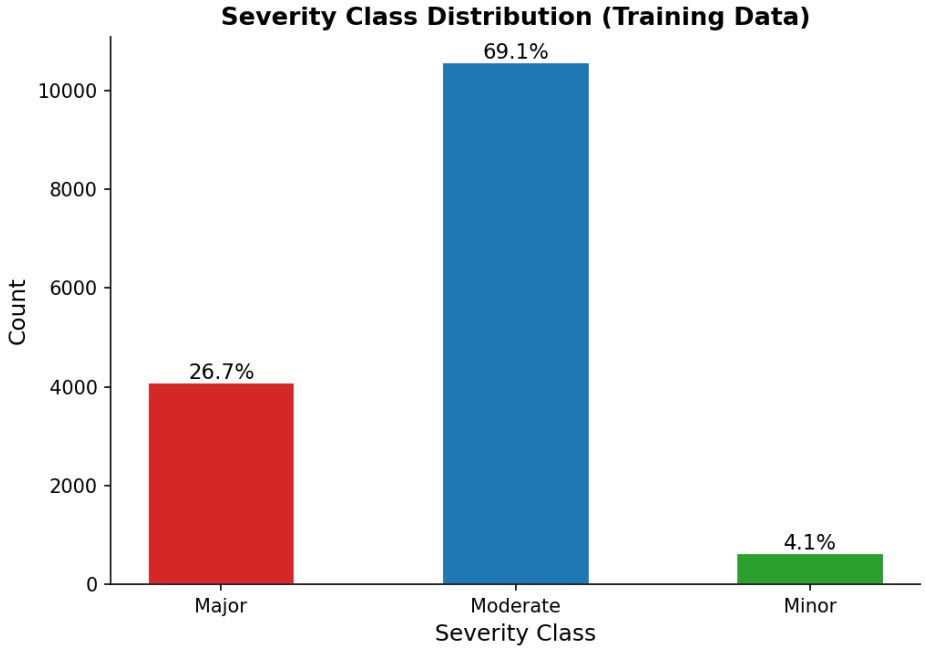
Adverse Event	RMSE (masked)	Inverse RMSE
General physical health deterioration	4.610	0.1783
Tremor	4.612	0.1782
Pneumonia	4.618	0.1780
Pyrexia	4.580	0.1792
Asthenia	4.781	0.1730
Oedema peripheral	4.787	0.1728
Somnolence	4.877	0.1701
Hypoaesthesia	4.917	0.1690
Weight increased	5.067	0.1648
Fall	5.162	0.1623
Back pain	5.213	0.1610
Haemoglobin decreased	5.353	0.1574
Hypotension	5.461	0.1548
Abdominal pain	5.464	0.1547
Decreased appetite	5.532	0.1531
Anxiety	5.510	0.1536
Cough	5.648	0.1504
Constipation	6.303	0.1369
Renal failure	6.162	0.1396
Dehydration	6.720	0.1295
Urinary tract infection	7.082	0.1237
Anaemia	7.291	0.1206
Drug interaction	7.544	0.1170
Hypertension	7.528	0.1173
Renal failure acute	9.544	0.0948
Atrial fibrillation	10.435	0.0875

Table B15. Estimated Composite Hardcore Clinical Score

Component	Value
Severity Macro F1 (x0.40)	0.5666
Side Effect Micro F1 (x0.30)	0.5772
PRR Inverse RMSE (x0.30)	0.2023
Estimated Composite Score	**0.4605**
Kaggle Submitted Score	**0.5134**

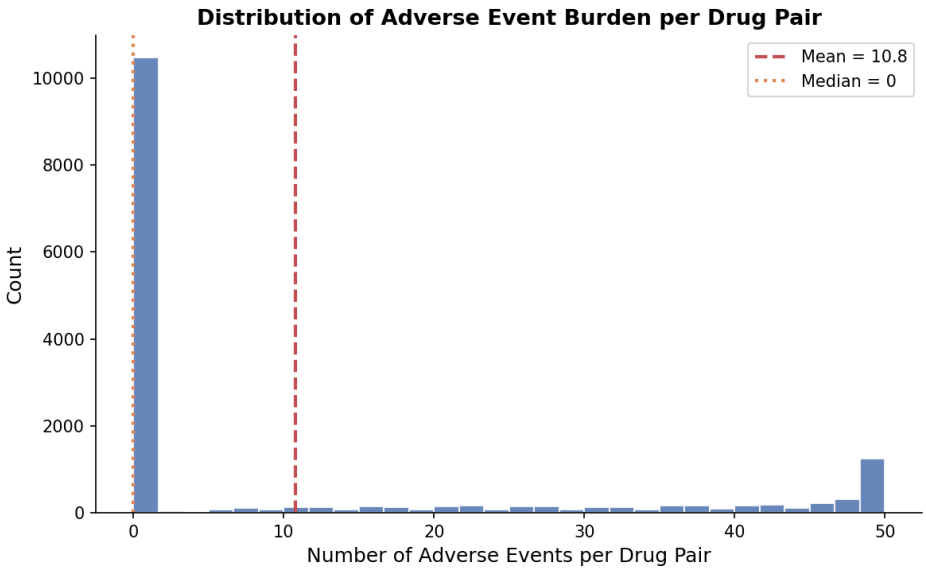
C. Figures

Figure C1. Severity Class Distribution



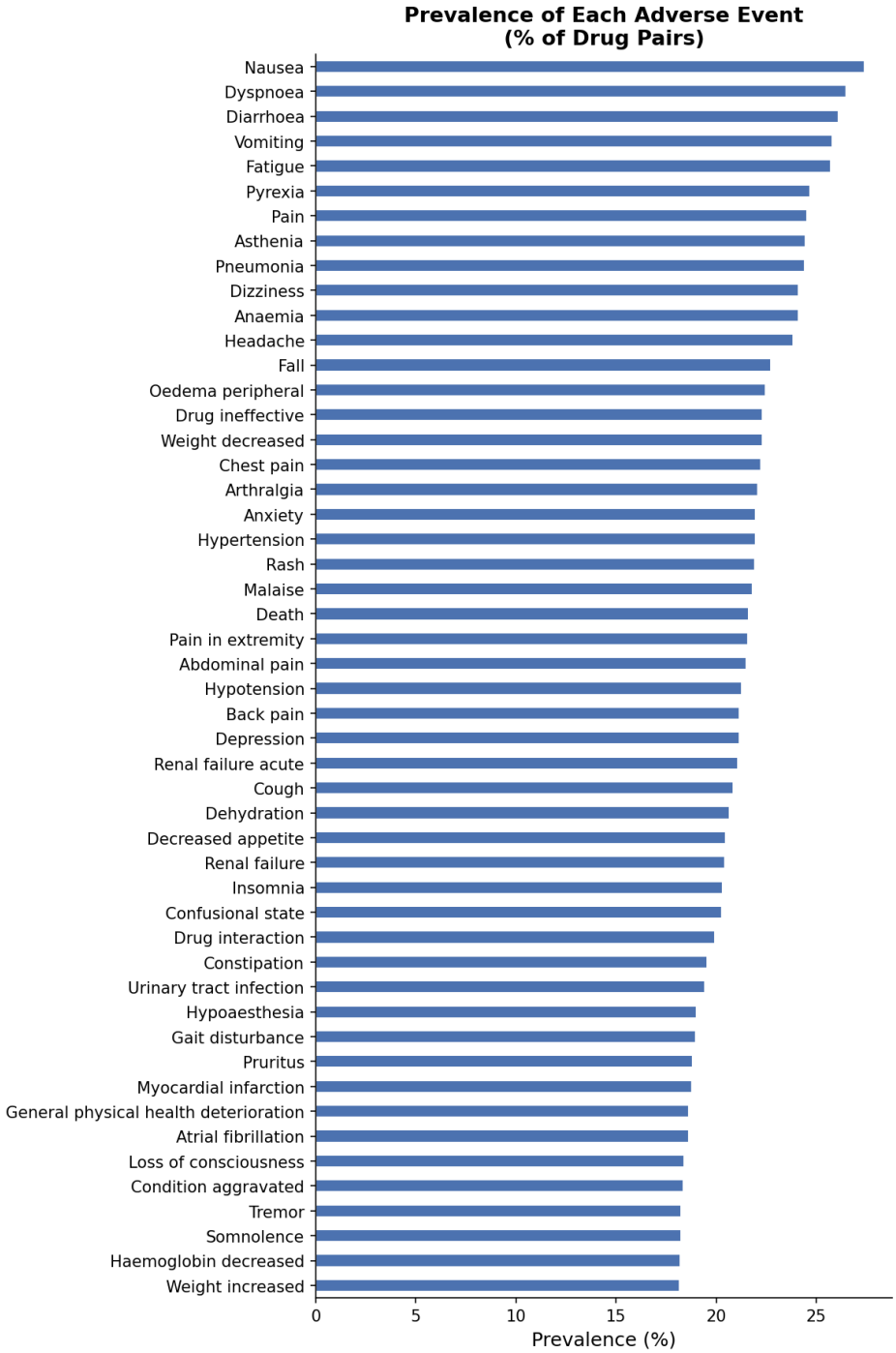
Severity Class Distribution

Figure C2. Adverse Event Burden per Drug Pair



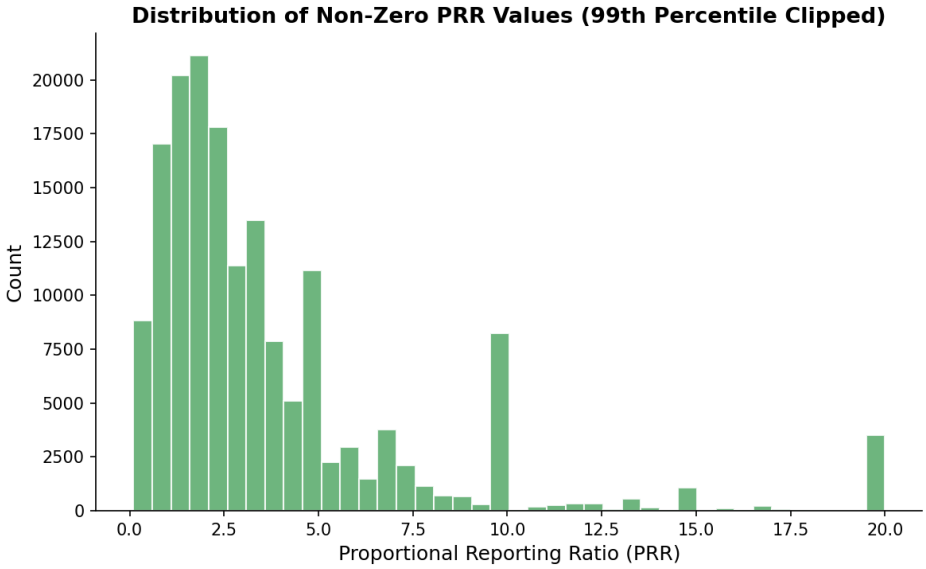
AE Burden Distribution

Figure C3. Prevalence of Each Adverse Event



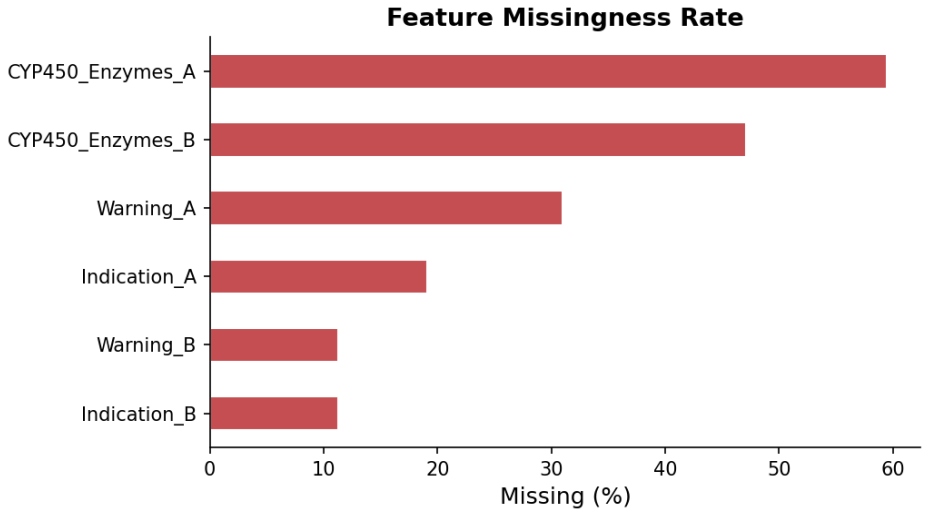
AE Prevalence

Figure C4. Distribution of Non-Zero PRR Values



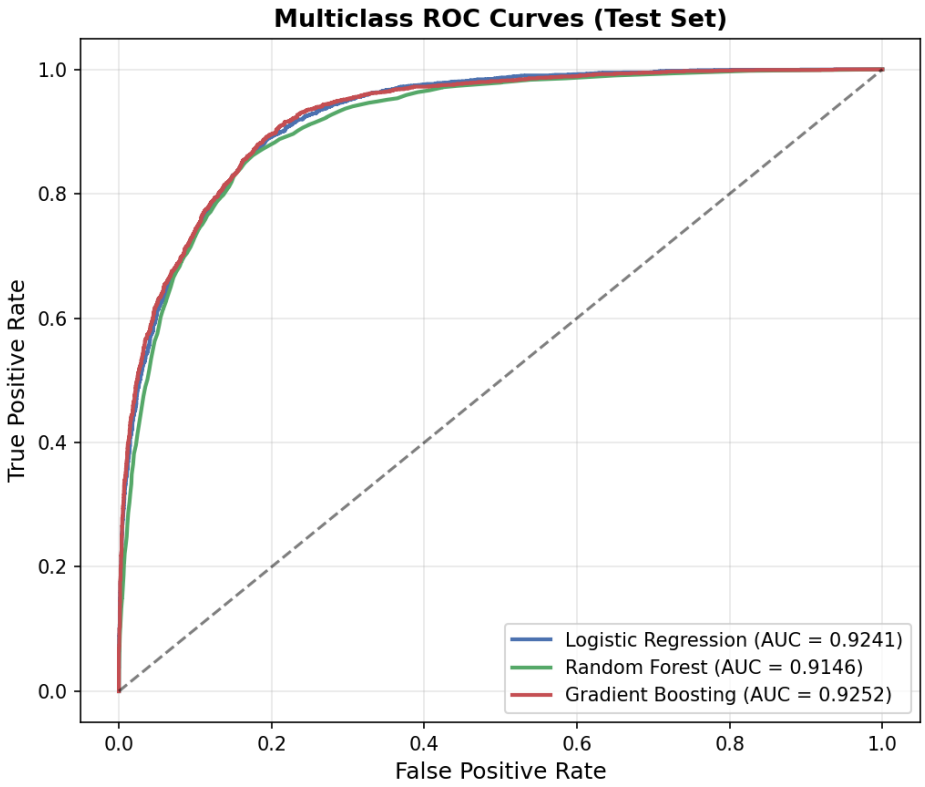
PRR Distribution

Figure C5. Feature Missingness Rate



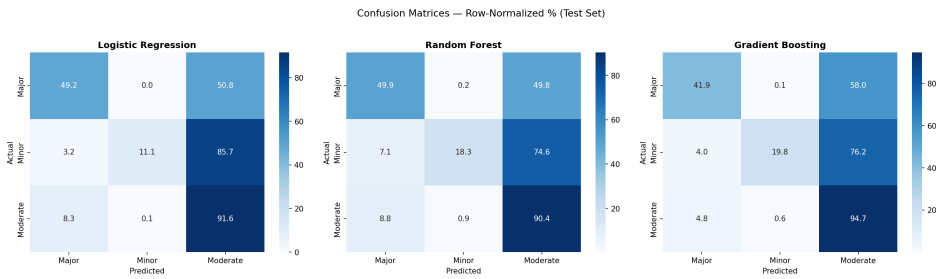
Missingness

Figure C6. Multiclass ROC Curves (Test Set)



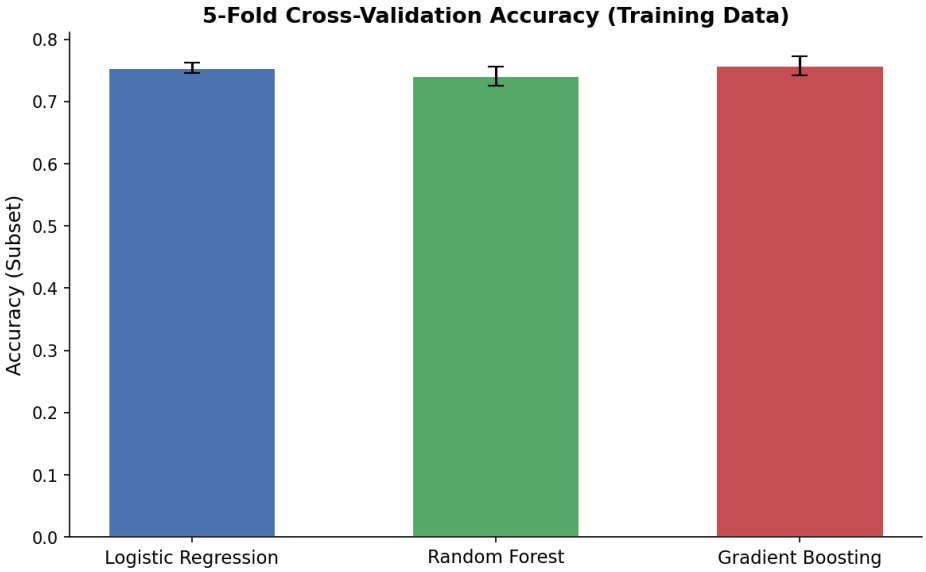
ROC Curves

Figure C7. Confusion Matrices - Row-Normalized Percent (Test Set)



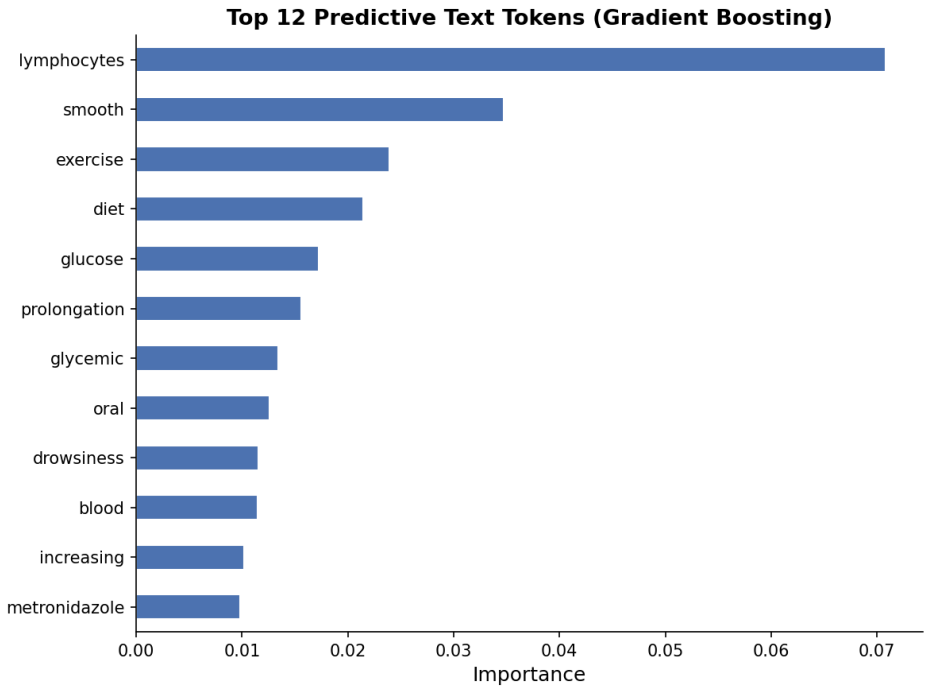
Confusion Matrices

Figure C8. 5-Fold Cross-Validation Accuracy



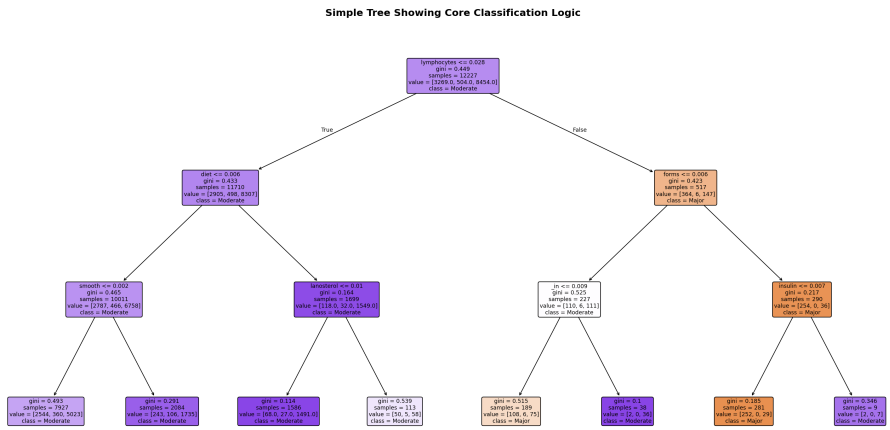
CV Accuracy

Figure C9. Top 12 Predictive Text Tokens (Gradient Boosting)



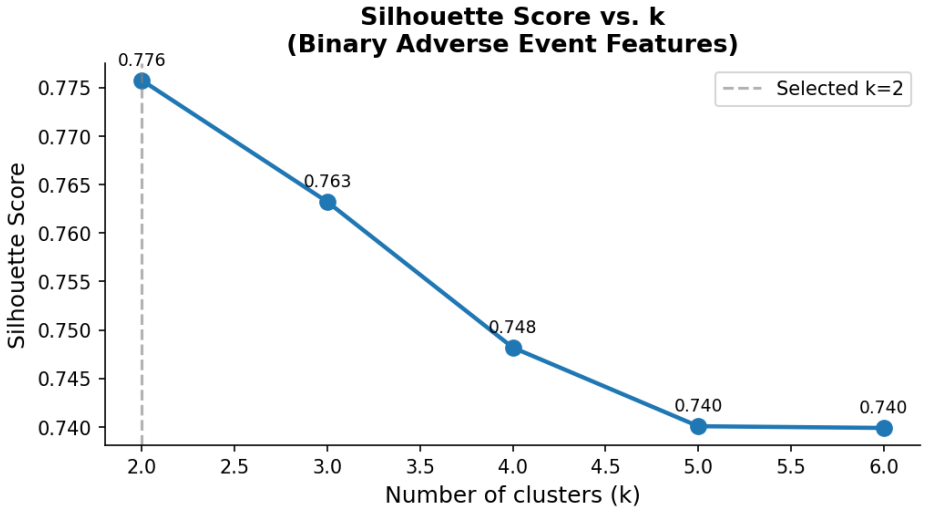
Feature Importance

Figure C10. Simple Tree Showing Core Classification Logic



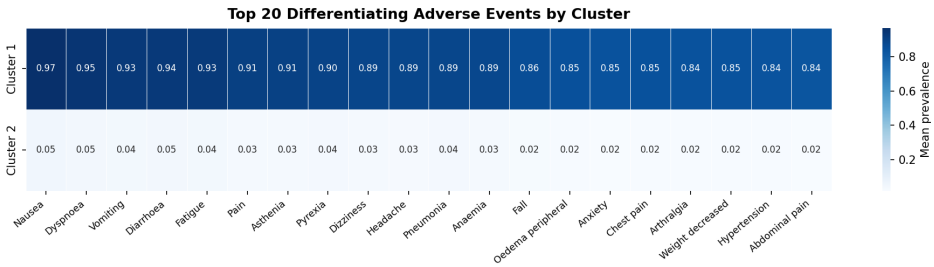
Decision Tree

Figure C11. Silhouette Score vs. k (Binary Adverse Event Features)



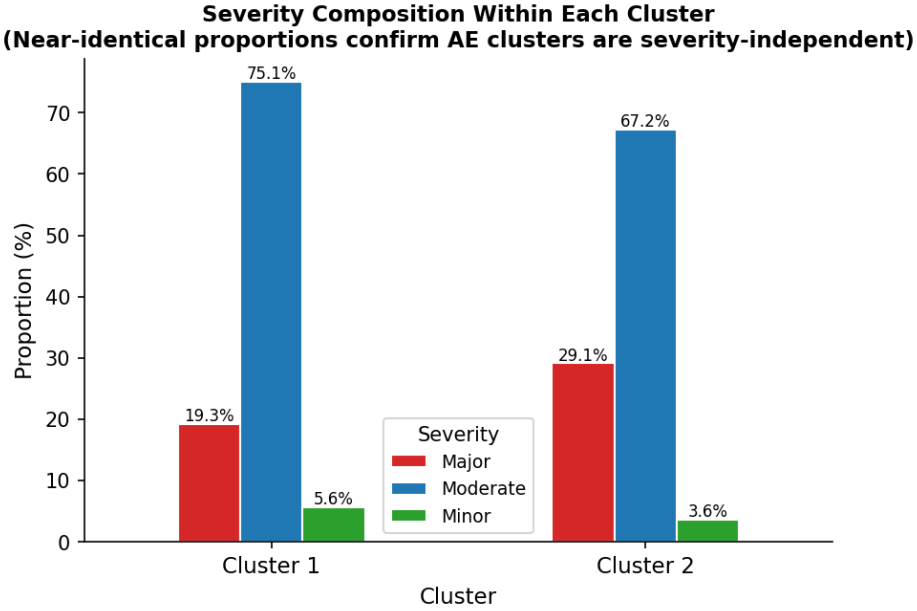
Silhouette Score

Figure C12. Top 20 Differentiating Adverse Events by Cluster



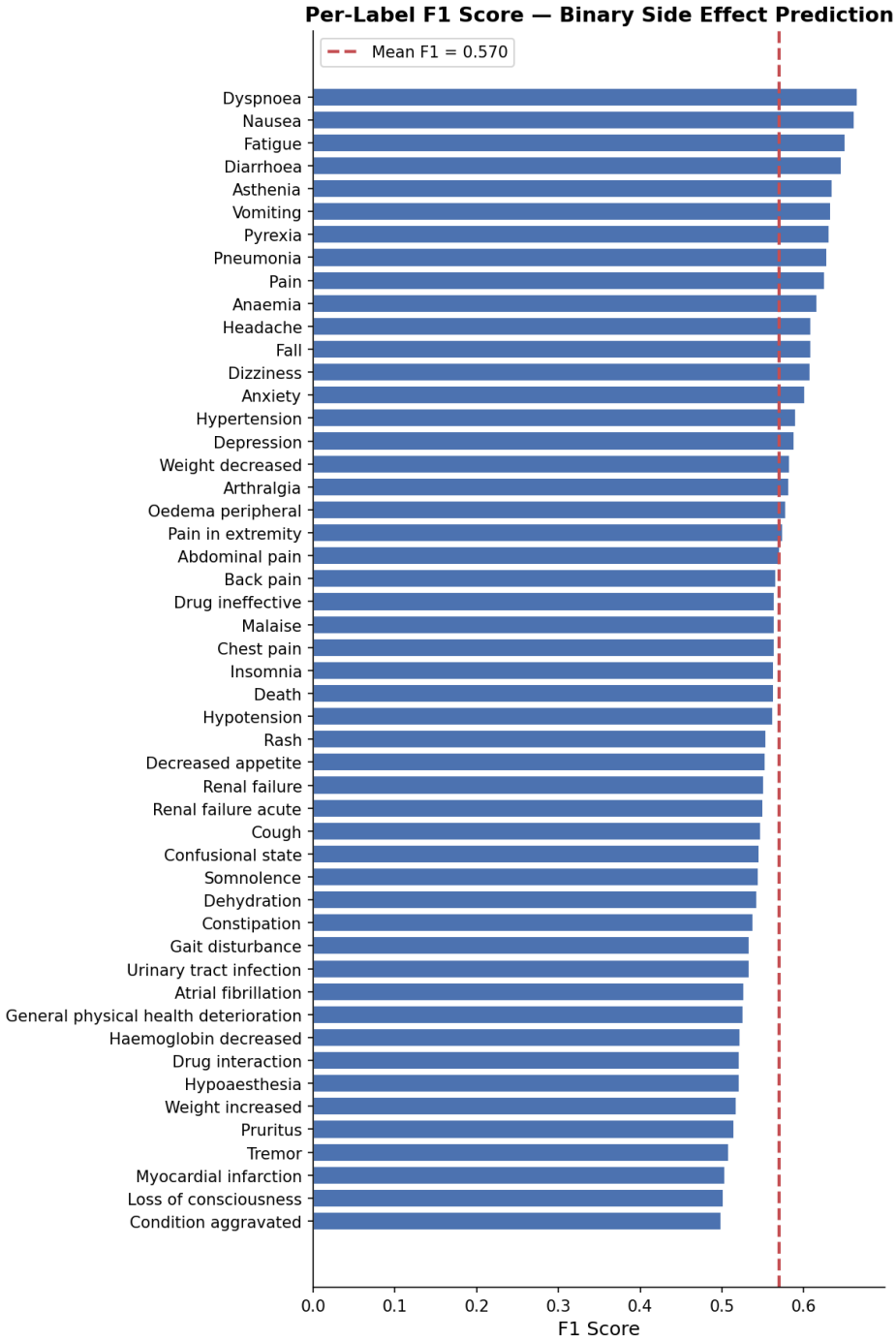
AE Profile by Cluster

Figure C13. Severity Composition Within Each Cluster



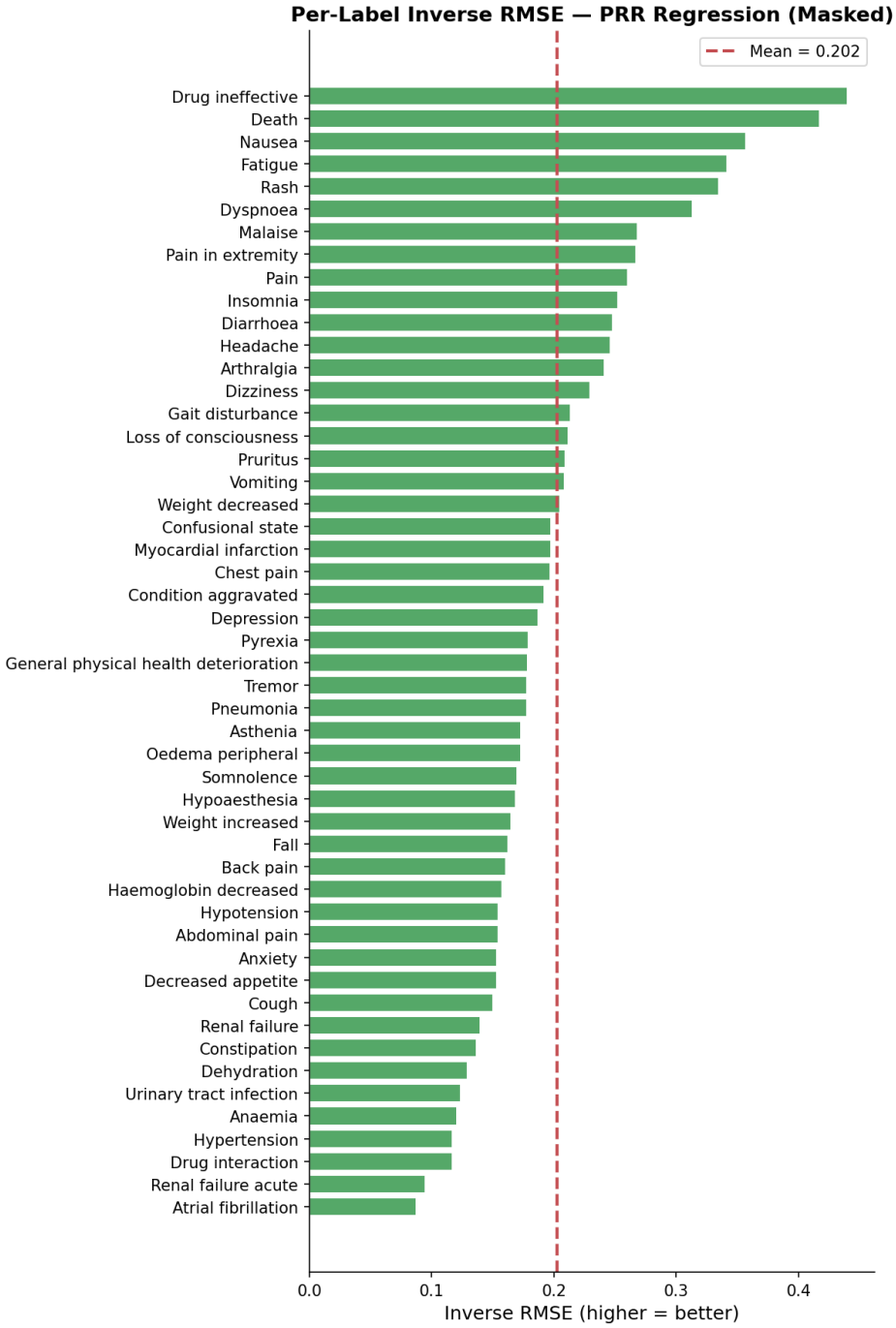
Severity by Cluster

Figure C14. Per-Label F1 Score - Binary Side Effect Prediction



Binary F1 Per Label

Figure C15. Per-Label Inverse RMSE - PRR Regression



PRR Inverse RMSE

Citations

1. Zhao D, Huang P, Yu L, He Y. Pharmacokinetics-Pharmacodynamics Modeling for Evaluating Drug-Drug Interactions in Polypharmacy: Development and Challenges. Clin Pharmacokinet. 2024;63(7):919-944.

2. Club Scientifique ESI, Hammani A. Deep-Pharma Multimodal Intelligence - DataHack 3.0 [Internet]. Kaggle; 2026 [cited 2026 Apr 15]. Available from: https://kaggle.com/competitions/deep-pharma-multimodal-intelligence-challenge-data-hack-3-0